

# Third Assignment: Primal–Dual Bounds and Branch-and-Bound Algorithm

## Instructions

- Answer conceptual questions clearly and precisely.
- Show all reasoning in computational parts.
- For algorithmic parts, provide:
  - A flowchart
  - Clear pseudocode

## 1 Primal and Dual Bounds

### Conceptual Questions

1. What is a primal bound in a minimization problem?
2. What is a dual bound?
3. Why is a dual bound always optimistic (for minimization)?
4. Define the optimality gap.
5. When can we certify optimality?

## 2 LP Relaxation

Consider the following integer program:

$$\begin{aligned} & \max 5x_1 + 4x_2 \\ & \text{subject to} \\ & 6x_1 + 4x_2 \leq 24 \\ & x_1, x_2 \in \mathbb{Z}_{\geq 0} \end{aligned}$$

1. What is an LP relaxation of an IP/MILP?
2. Solve the LP relaxation of the given example using geometric approach (geometric approach is a very simple approach and you can find it in almost every standard operations research/linear programming text book).
3. Use it to compute a dual bound.
4. Find a feasible integer solution and compute a primal bound.
5. Compute the optimality gap.

## 3 Duality in Linear Programming

### Conceptual Questions

1. What is the dual problem of a linear program?
2. State the Weak Duality Theorem.
3. What does Weak Duality imply about bounds?
4. State the Strong Duality Theorem.
5. What is complementary slackness (conceptually)?

### Exercise

Consider the following LP:

$$\begin{array}{ll}
 & \max 3x_1 + 2x_2 \\
 & \text{subject to} \\
 & x_1 + x_2 \leq 4 \\
 & 2x_1 + x_2 \leq 5 \\
 & x_1, x_2 \geq 0
 \end{array}$$

1. Write the dual problem.
2. Solve the primal geometrically.
3. Solve the dual.
4. Verify strong duality.

## 4 Bounding Techniques

### Conceptual Questions

1. Why does LP relaxation provide a valid bound?
2. What is a Lagrangian relaxation (conceptually)?
3. What other general (problem-agnostic) methods can you list with which one can obtain a dual bound for their given problem. It is enough to list a few and describe them.
4. What makes a bound “strong”?
5. (Optional question; you may answer it or not): compare the discussed approaches with each other in a table considering use cases, i.e. LP, IP, and MILP. Try to include properties.

### Exercise

For the knapsack problem:

$$\begin{aligned} \max \quad & \sum_{i=1}^n p_i x_i \\ \sum_{i=1}^n w_i x_i & \leq W \\ x_i & \in \{0, 1\} \end{aligned}$$

1. A greedy algorithm can solve the LP relaxation of the 0/1 Knapsack problem. Describe how it works and write its pseudocode.
2. Explain how the fractional knapsack solution provides an upper bound.
3. What is integrality gap?
4. Compute the dual bound for the following using the greedy algorithm:  
Profits:  $p = (10, 8, 6)$  Weights:  $w = (5, 4, 3)$  Capacity:  $W = 7$

## 5 Branch and Bound; An Exact Algorithm

### Conceptual Questions

1. What is an exact algorithm? When it is used?
2. What is the main idea of Branch and Bound?
3. What are the three main components?
4. What is fathoming?

## Exercise

Solve the following problem using a Branch-and-Bound tree:

$$\max 3x_1 + 2x_2$$

subject to

$$2x_1 + x_2 \leq 4$$

$$x_1, x_2 \in \mathbb{Z}_{\geq 0}$$

1. Draw the full search tree.
2. Compute bounds at each node.
3. Indicate when nodes are pruned.
4. Identify the optimal solution.

## Flowchart and pseudocode

Draw a flowchart and write a pseudocode for the generic Branch-and-Bound algorithm for maximization problems.

Your flowchart must include:

- Initialization
- Node selection
- LP relaxation
- Bounding test
- Branching
- Termination